

IMPRINT



VOLUME II ■ DEPARTMENT OF ZOOLOGY ■ XZA ■ 2017 ISSUE



Cover Design: Siddharth Bhatia

OF PLATO AND PLATYPUS

THE XAVIER'S ZOOLOGY ASSOCIATION



Members of the Xavier's Zoology Association

The Xavier's Zoology Association established an year ago is at the forefront of running exploratory activities of the Zoology Department for developing interest of students in the subject of zoology and nature in general. Imprint is inclusive of one of the initiatives of the association to develop interest in the field and provide information about the subject to a wider audience. Here's this year's student team for the magazine!



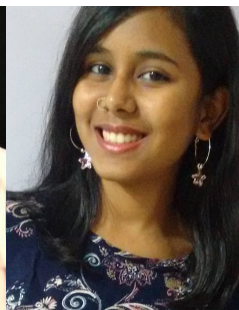
HARSHIT
SINGH



SHREYA
DIMRI



ZACHARY
BORTHWICK



ISHA
SHYAM

LETTER FROM THE EDIT TEAM

Dear Readers,

Human and animals have been related to each other for a long while. Whether in conflict or in synchrony, the situation has stood for as far as human memory can stretch back. And why would it not be? For man after all is indeed an animal for all taxonomic purposes. The interactions between man and animal are shrouded in misrepresentation. Had there been not a lot to relate, why would poetry of Walt Whitman talk of dallying eagles and that of Lord Tennyson of a brook and its creatures – all in the context of relating humans and the wild?

In the second edition of Imprint this year, the theme - 'Of Plato and Platypus' aims to bring you to a deeper understanding of these relations and how they have developed in the world as we know. Plato in philosophies stood just as much an oddity to the human world as a Platypus is in that of animals. Comprehending the human - animal relationship is much like understanding them two and we hope our stories help deliver the same in this issue of the magazine!



Sujata

Dr. Sujata Deshpande
Editor - in - Chief

Harshit

Harshit Singh
Student Editor-in-Chief

Shreya Dimri

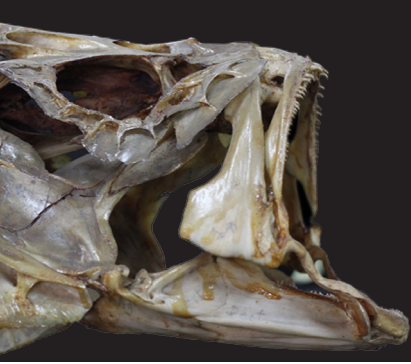
Shreya Dimri
Design Editor

Zachary

Zachary Borthwick
Content Editor

Isha

Isha Shyam
Content Editor



Xavier's Zoology Association Activities



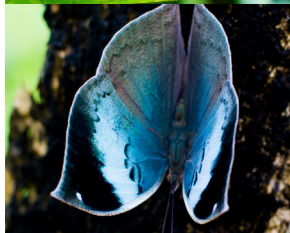
The Walk Of Life

To kick off the association's activities for the academic year 2016-17, the Department showcased its best and rarest specimens to all students and staff of the college on the 12th of August 2016. Guests got a chance to take the walk of life through the evolutionary ladder in under 10 minutes by passing from exhibit to exhibit. Exhibits ranging from

Phylum Protozoa, to the extensive Insecta collection that the department boasts off, to class mammalia were displayed to over 400 people who entered our Zoology lab. With such an overwhelming response the XZA was ready to bring to the college a line-up of even more exciting events.

A Visit To Ovalekar Wadi

Butterflies are the only insects you probably won't be squeamish about, what with their fantastic colours and vibrant patterns. The association organised a field trip to Ovalekar Wadi, Thane where Soham Pattekar- an alumnus of the department guided an eager bunch of future Zoologists and butterfly enthusiasts along the trail. Stopping occasionally and imparting knowledge of these pretty winged creatures.





Paws For A Cause

Wagging tails and barking sounds echoed through the foyer when Shirin Merchant returned to her alma mater with puppies who care. She spoke about the amazing ability of dogs to be used in play therapy, search and rescue and assistance to the differently abled. She also enlightened eager minds on how they can take it up as a profession. The rest of the event was spent in some very important human-canine bonding. As the Dogs enjoyed the attention and the students just couldn't get enough of their cuteness.

Blood Lions

This hard hitting documentary about the reality of the conservation efforts for lions and other predators in Africa was screened by the association on the 14th of January 2017. The documentary attempts to uncover the malpractices like inbreeding from the blanket of conservation and research. Following this the students and teachers engaged in a discussion with Pippa Hankinson- producer of Blood Lions, Bittu Sehgal and several others.



Career Pitch

The association had the privilege to host Dr. Y.V. Jhala as he gave a talk on 'Technology Applications and Careers in Wildlife'. Dr. Jhala gave his speech to a room full of budding zoologist and gave them an idea of what it's like to work in the field of conservation by telling them about the duties of a conservationist and the tools at his/her disposal. As conservation is the need of the hour we are glad Dr. Jhala could come and speak to us and encouraged the young minds listening to him to take up conservation and become warriors for the planet.

Of Plato & Platypus

Man and animal have lived beside each other for ages, both in synchrony and in conflict. What is it that essentially binds the two and what is it that brings them together?

In the sphere of life, how far do we even stand away from each other? For that starts a time for Plato and Platypus to sit down and have a chat.

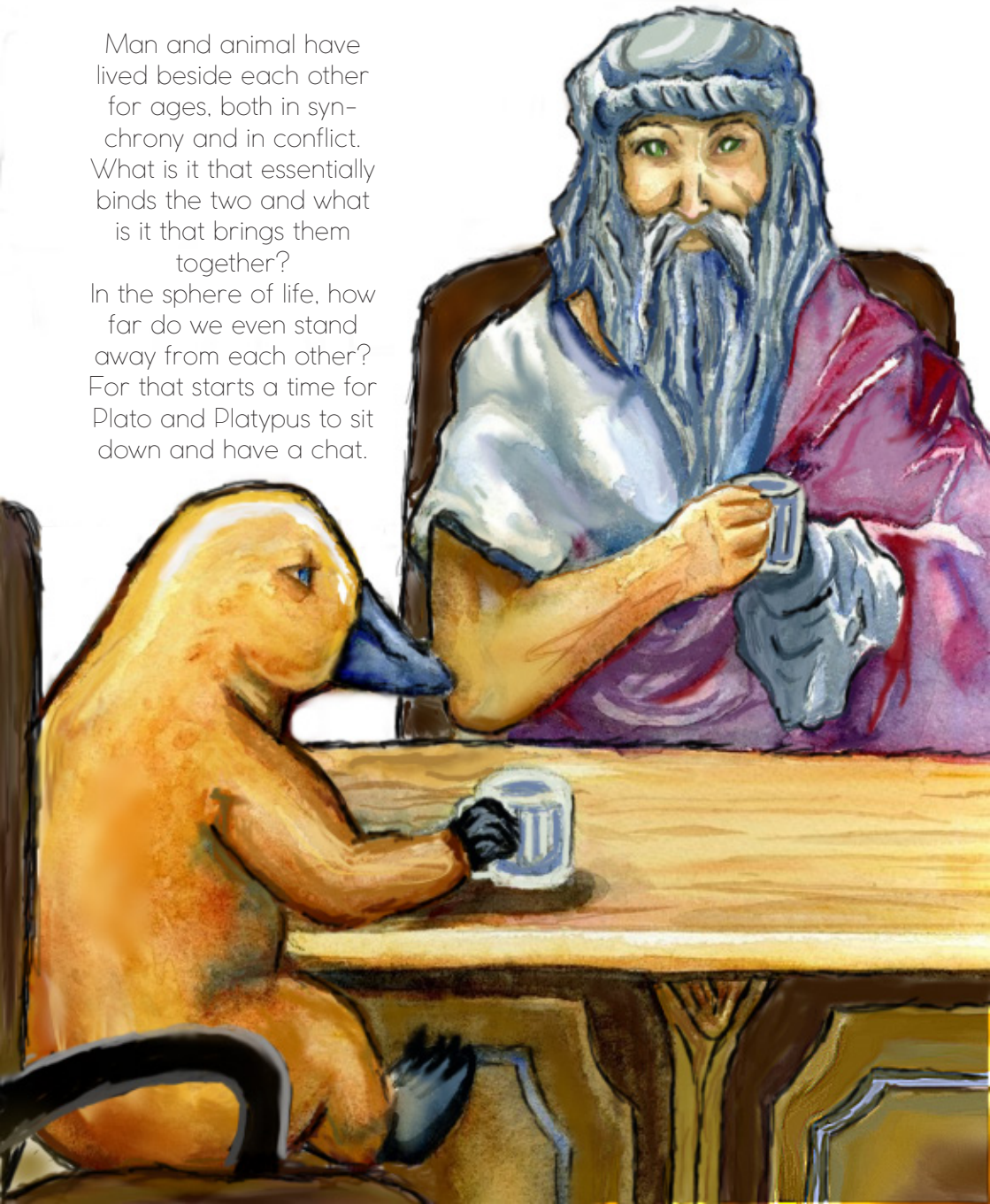


Table of Contents

EXCLUSIVE:

**Feature
your pet**

Pg 4

Pawsitive Therapy 01

Of Flying Tigers 05

Unreal Debate 08

Conflict & Understanding 11

A Tern Tale 15



In Conversation With..
Dr. Claire Spottiswoode 17

If The Crocodiles Go, We Go 21

The Spirits Within 23

Like The Mad Hatter's Tea Party 32

Internships Galore 35

The Strange Existence of
Velvet Worms 37

A woman with curly brown hair and glasses is smiling and holding a long-haired, golden-brown dog. The dog is looking up at her. The background is a plain wall.

PAW'SITIVE THERAPY

Dinesh* had a dream. A strapping young man, he had joined the army with the aim of defending his country, and marrying off his teenage sister. Sadly, in the blink of an eye and the sound of an explosion, the dream turned into a nightmare. He awoke one day to find himself in a hospital bed, his legs replaced by bandaged stumps. "He has been depressed ever since he gained consciousness. He is so ashamed of his condition that he refuses to meet his family and has tried to take his life on many an occasion," the dean spoke. I watched as Dinesh glanced over at the small black Labrador- Honey, by my side.

Honey and I were on a visit to the army rehabilitation home, in Pune, to demonstrate the benefits of Assistance Dogs to the men crippled by the Kargil war. As the dean continued about how Dinesh hardly ever spoke and hadn't smiled for the past six months, Dinesh wiggled his fingers at Honey. The Labrador in her needed no further invitation. Tail wagging furiously she ambled over and placed her orange squeaky ball right on his chest and stepped back, her eyes inviting him to play. Dinesh was delighted! The corners of his eyes crinkled as he made an enormous effort and swapped the ball to the floor.

Honey, equally thrilled at finding a playmate, grabbed the ball and promptly deposited it back on his chest. They played the game for half an hour. When it was time to leave, Dinesh encouraged Honey up onto his bed and gave the panting Lab a huge hug as tears rolled down his face onto her black fur.

Honey, a gentle dog rescued from an abusive owner, has been repairing broken hearts and healing wounded souls for the past ten years. Till date, she hasn't failed once to reach out and cheer up a depressed person. She adores people, just loves to be around them and most people, especially children, love to interact with her. They respond to Honey in ways that they can't and don't respond to other humans.

Ever since man befriended the wolf and converted it into *canis lupus familiaris*, dogs have been man's best friend. Drawings found in caves in Bhimbetka, Central India, depict a man walking a dog on a leash – these caves were supposedly inhabited by man over 100,000 years ago!

Ever since then, man has been realising new ways that dogs enrich our lives - from therapy dogs, like Honey, to assistance dogs for the disabled, guide dogs for the blind, sniffer dogs that can sniff out anything from rhino tusks to drugs and even cancer!

Therapy animals are especially useful to children with behavioural disorders, or those who have been physically or emotionally abused. Iqbal* was one such child. The first time I met him he was aloof and avoided human touch. Initially, he watched impassively as the other kids petted Honey. It took all of ten minutes and a lick from Honey to bring a smile to his face. Soon he was playing fetch with her and feeding her biscuits. With Honey, small tasks became fun and inspired Iqbal to improve his hand-to-eye coordination. Honey didn't mock if he couldn't tie his shoelace. She waited patiently till his fumbling hands finally tied the knot. A dog pays little attention to age, looks or disabilities and accepts people as they are, which is why children, like Iqbal, show marked improvement in canine-assisted therapy sessions.

But the job for dogs doesn't end there. Sanam Karunakar was a normal teenager involved with college and other activities, looking forward to a promising career. One unfortunate day she was involved in a car accident, which claimed both her parents and left her paralysed from the chest down. Life seemed meaningless to her till she met a rambunctious Labrador puppy - Magic!

"I had resigned myself to this dependant lifestyle, till I met Magic," reminisces Sanam. "Until then I had only seen Assistance Dogs on television and would have loved to have one but I was not aware that this was possible in India. Living with a disability is an endless struggle to perform even the simplest of tasks. For most people, dropping a pen is an inconsequential event. When you are seated in a wheelchair, with limited mobility, the pen is often out of reach. Being forced to constantly ask for assistance of others is depressing, especially when I used to be so active before."

Magic stayed by Sanam's side all through his life and helped her with daily tasks such as opening doors, picking up dropped items, calling for help, retrieving the



telephone and even covering her with a blanket. "The day Magic entered my life the world became a sunnier place to live in," says Sanam. "He was my companion when I was alone, my laughter when I was sad and most importantly he was my best friend. Magic loved me unconditionally. He brought back confidence and hope, once lost, and since the day I got him, I knew nothing was going to be the same again."

Every dog owner knows how much unconditional love and happiness a canine family member provides. In fact, studies have shown that people who own dogs and other animals are less stressed out, have fewer health problems and are generally happier people. Sigmund Freud, the father of psychoanalysis, was the first to emphasise the therapeutic value that dogs offer people. Since then, mental health experts the world over have recognised the role dogs can play in comforting, calming and soothing the suffering.

Therapy dog work is mostly done gratis all over the world. The best reward comes in the form of the smiles from people who haven't smiled for months, or the tears of gratitude from people who have been able to let go of emotions that have been pent up for years on end. And as we move into a new year, we realise the truth in Roger Caras' words-

"Dogs are not our whole life, but they make our lives whole."

Honey, Magic and many other amazing working dogs have long since gone to a happier place; but their work, their love and their belief in helping unconditionally will live forever in the hearts of those they touched.



Shirin Merchant an ex Xavierite, is Asia's only internationally accredited dog trainer and behaviourist. She runs the Canines can Care organisation and trains assistance dogs, of which we had a first hand experience recently when she delivered a talk on campus.

* Names in the article have been changed to protect privacy of the given individuals mentioned about here,

FEATURE YOUR PET!

We asked you for them and here are the ones who really made it all the way to being featured here with us!!!



SMOKEY

We adopted Smokey from a shelter in Pune 9 years ago. He had been abandoned and horribly mistreated. Over the years, he's now a pampered "favourite child" of a Naval dad and mum, and therefore has travelled much of India. From beaches to mountains, boats to trains, he has done it all, and while he enjoys the outdoors on holiday, at home he'd rather sleep till 11!

- Lisa Gonsalves

Bolt. That is the name of my pet Labrador or rather I should say my little brother which is more apt. He is the very embodiment of love, trust, strength, patience, understanding, bravery, knowledge, looks, you name it. All the very best virtues you can think of. He's made it through *Puppy Strangles* and is an amazingly brave dog.

-Keren Pereira



BOLT



PRESTO

The name Presto is very dear to our family as my mom's three pets since childhood were named Presto... Like his name, his habits are also quite unique, he has a habit of sleeping on the tea table and he daily sleeps on my bed... My pillow... He wouldn't sleep till everyone is back home. Presto is like a part of our family and is always running and jumping!

-Archie Joshi

Of Flying Tigers...



As time has set foot forward, urbanization across the world has been growing rapidly . At an incredibly fast rate, 66 percent of the world population is projected to be urbanised by 2050 (UN 2014). What has come along is not only the loss in agricultural and forest landscapes, but also in biodiversity of these urbanised scapes. An inadequacy in the understanding people have about the ecological consequences for urbanization gives rise to the challenge of regulating city planning manifold. This brings to attention the fact that despite changing landscape, urban environments still offer multiple opportunities to organisms, especially in the tropics. It becomes important to study the native habitat refugium that is presented in cities for species that are caught in the thralls of urbanization or perhaps are even attracted by it. It is not even a question as to why suddenly ecologists have a piqued interest in studying interactions within the urban sphere of life, for it indeed matter in the times ahead. Research in urban tropics has had its major problems. Lack of studies in tropical zones (with higher biodiversity) with even fewer studies on top vertebrate predators in the matrix, coupled with studies foregoing the idea of being able to cover the whole gradient of urban setting speaks volumes of this neglect. This restricts understanding of urban adaptability, which is key to the working relations with urban organisms in their novel habitat – a major lacunae in the research.

Our work at the Black Kite Project in the National Capital Territory is precisely about filling in these gaps by trying to understand the ecology of a major avian predator in an urban setup. This provides for a model

raptor research while underlining interactions between kites, people and the urbanscape in which they coexist. With monitoring of over 200 nests since 2012 in the urban space of Delhi, a city second only to Tokyo in its urban expansion rate, the project aims at building an understanding of how such massive populations of a raptor sustain themselves in a city like Delhi and why does it work at all for a specie with a wingspan of more than four feet? Record keeping involves intensive and meticulous work of measuring, banding and tagging nestlings and adults alike, and it takes not just research scholars but a truly skilled team of field assistants who risk their lives to make that happen!

Black Kites or *Milvus migrans* seem to be the most unlikely of birds in an urban habitat. There have been various speculations among people about how the kites have come to be this successful, including the suggestions around occupation of the niche once covered by the now locally extinct vulture species of India, the White Backed Vultures *Gyps bengalensis*. However, the research findings over the years have shed light upon much more, including the differing races in the population of kites in Delhi, that includes the smaller *govinda* sub-species (also called Small Indian Kite) and the larger *lineatus* (Black-eared Kite) with the latter migrating all the way from Central Asia. They also have interesting behaviours which go beyond simply being dictated by a desire for food, such as sun basking whenever they seemingly have the time for it.

Black Kites or *Milvus migrans* seem to be the most unlikely of birds in an urban habitat, there have been various speculations for the same though, including looking at possibilities as to how they might have replaced the ecological niche in India as once covered by Vultures.

The birds however, are not completely independent in the urban zones and get subsistence to a very large extent by humans themselves in Delhi. Philanthropic meat tossing is popular amongst a lot of Islamic communities as a daily custom and is well practiced in the city. Poultry refuse too is open to these raptors to snatch up at a variety of locations across the city, whether they are large butcheries or small mandis. Hence, a primary requirement is well looked after for within the urban zone. Highly dense colony pockets of kites occur in the city frequenting trees as nesting sites and the presence of human subsidy in terms of food pushes for habitat selection at varying levels, from coarse to fine, shedding



light on how complex ecology can be within an urban setup. Differently developed parts of a city provide for unique urban pockets, which may or may not favour an organism's presence or offer a niche position. The city itself is not a homogenous patch for an individual organism and the preferences a species makes about inhabiting a certain locale can vary. This is important while considering a population constantly well encountered across most areas in the city, and throughout the year. An area or a niche that a species may inhabit in an urban ecosystem may highly depend on how the species came to be there in the first place. They may be synanthropic species that take refuge within the remnant habitats in an urban scape or on can be synurbic and have extended adaptability to live in an urban environment, shown usually by markedly higher population densities in the urban pockets.

“...not just survive but thrive along with people...”

Black Kites, being a top trophic level organism, appear to be a key species to understand how urban expansion can possibly be directed in a manner for maximum efficiency towards catering to a decrease if not a halt for biodiversity loss. With a promise of an enormous sample sizes to science, species such as these are true hope bearers in understanding the ways we can create a sustainable future.



Urvi Gupta is the Assistant Research Manager at the Black Kite Project and is currently pursuing her Masters from The University of Oxford. Harshit Singh was a student volunteer for the Black Kite Project.

The Unreal Debate

Are we shaped by our experiences and upbringing,
Or do our genes dictate what we can become?

For centuries philosophers and social reformers tried to solve the riddle of “nature versus nurture” but only relatively recently has scientific evidence been brought into the debate. The biggest question we are trying to answer is are we shaped by our experiences and upbringing, or do our genes dictate what we can become? In simple words the phrase ‘nature vs nurture’ relates to the relative importance of an individual’s innate qualities i.e sense of nativism or innatism as compared to an individual’s personal experiences i.e sense of behaviourism or empiricism.

Tabloids like The Daily Mail has made headlines because of articles like “Were you born to stray because you carry an infidelity gene?” The implication was obvious: a woman who keeps having flings just can’t help it- it’s in her genes. This isn’t completely true since research shows genes do not compel behaviour and that infidelity is likely to depend on a host of other factors in any case. The debate between nature vs nurture is never ending because every now and then articles come linking certain set of behaviour with genes and which implies we all are in thrall to our genes.

The English philosopher John Locke stated the “blank slate” view for human behaviour which basically means that the human mind has no innate traits, so that their behaviour, abilities and personalities are entirely the result of interaction with their environment. This view has led to bizarre theories of child-rearing and untold guilt among parents who blame themselves for the failings of their offsprings. John Locke also believed that he was striking a blow against repressive concepts such as original sin and divine right of kings.

If humans were all born equal then everyone could and should have as much right to life, liberty and happiness- a view that impressed Thomas Jefferson, principal architect of the US Declaration of Independence.

Tabloids like The Daily Mail has made headlines because of articles like “Were you born to stray because you carry an infidelity gene?” The implication was obvious: a woman who keeps having flings just can’t help it- it’s in her genes.

Herbert Spencer and Francis Galton tied Darwin's theory of evolution to the study of human society. Spencer came up with the summation of evolution as "survival of the fittest" and Galton came up with concept of "eugenetics" - the systematic "improvement" of the human race through selective breeding. Galton followed up the cases of identical twins since they were the staple of nature vs nurture debate and eventually concluded that nature prevails over nurture because of the many similarities found between identical twins. Leonard Darwin, son of Charles who was the first president of the Eugenics Society warned darkly of the threat to future generations of allowing the "unfit amongst men" to breed.

In the US, the concept of eugenics spread like wild fire and as a result sixteen states had laws permitting compulsory sterilisation of the "feeble minded". This followed in Germany by Nazis introducing sterilization of thousands with traits like schizophrenia and ended with slaughter of millions in camps.

In the US, the concept of eugenics spread like wild fire and as a result sixteen states had laws permitting compulsory sterilisation of the "feeble minded". This followed in Germany by Nazis introducing sterilization of thousands with traits like schizophrenia and ended with slaughter of millions in camps.

The American psychologist John Watson had proclaimed that all talks of traits and instincts are beyond quantification and thus meaningless. Winthrop

Kellog studied the role of environment by putting an ape with his child. B.F Skinner put his daughter in specially constructed box for several hours. Watson himself conditioned his child. The problem with these experiments is, though they claim that the environment was taken into account, the genetics can't be ruled out completely.

One of the landmark experiments were done by psychologist Harry Harlow. He separated baby monkeys from their mothers and put them in cages containing two artificial mothers. One was just a wire fitted model fitted with a feeding bottle and milk, while the other was more life like and cuddly- but



unable to supply milk. According to behaviourists, the monkeys should have quickly learned to ignore the coldness of the wire “mother” and got on with the vital business of taking her milk.

Yet Harlow found the monkeys spent most time with the cuddly but milkless mother, making only brief dashes to the wire version when hungry. Harlow had demonstrated what most people except behaviourists – would regard as obvious: behaviour is shaped by more than just the environment. The monkeys had an inbuilt instinct for what to expect from a nurturing parent and sought it out.

Among academics, however the debate rumbled on. In 1975 the Harvard expert E.O Wilson showed that genes can alone produce spectacularly complex behaviour. He also attempted to extend his argument to human behaviour and provoked a storm of controversy because it seemed to resurrect genetic determinism. Meanwhile, scientists continue to claim evidence for genetic influence on everything from sexual orientation to career choice.

By the mid 1990s the academic world finally appeared to be coming around to the same conclusion as the public: that human behaviour is a mix of nature, nurture and simple happenstance. The studies of human genome uncovered genes whose actions affect the ability to interact with others-making a mockery of the standard dichotomy between genetic and environmental factors.

After a century and more of claim and counterclaim, the nature vs nurture debate is finally being seen for what it is: an object- lesson in the dangers of thinking one side of any debate has a monopoly on the truth.



A leopard is walking from right to left in the foreground. In the background, a city skyline is visible at night, with many lights glowing. The leopard is in a natural, rocky environment with some dry grass and sticks.

Conflict and Understanding

How do we deal with Animals in our Spaces?

Humans have a very dichotomous relationship with wild animals. Some groups of people are scared of them while others respect or even worship them. Many wild animals in Indian and other traditional cultures are powerful symbols. Sadly today what we hear mostly are the negative interactions of what biologists call “conflict” which is dominates news reporting of wild animals today.

Although the term human-animal conflict is commonly used, we are essentially dealing with an issue of humans not expecting the wild animals to be in their space while wild animals do not understand man made boundaries and inhabit any area which best suits their requirements of resources like food, water and shelter. This phenomenon of negative human wildlife interaction is not restricted to India alone. Some European countries have killed and exterminated large carnivores found in their countries as a threat to human life or game. These very countries are now running programs of reintroduction of large carnivores due to past extermination programmes. Generally the two kinds of losses incurred due to wild animals are loss of livestock and crops. On rare occasions, losses of life or injuries to humans occur as well. We assume that carnivores should remain in the protected wilderness areas. But wild animals do not understand or follow man made boundaries and for a carnivore there is ample amount of easy prey available for it in human dominated areas. This is especially true in India with one of the highest density of livestock in the world.

Maharashtra itself has quite a few examples of carnivore existence close to human habitations. We had conducted a radio telemetry study on a tigress in Chandrapur District and the study shows that this particular tigress was living very close to humans on everyday basis, resting during the day in some nearby forest patch and active during night-time hunting domestic animals when people would be fast asleep.

Similarly another study from Ahmednagar also showed large carnivores such as leopards and hyenas living in sugarcane fields. These animals also followed a similar pattern of being active during the night and hunting domestic prey. In some parts of India we observe retaliatory killing of wild carnivores by people in response to losses incurred. Different cases of animal cruelty have been observed throughout India where people have burnt a leopard live once captured in a cage or killed a leopard by pelting with stones or sticks. Very often media publicizes negative stories. But these incidences majorly occur because there is lack of awareness that the people have about the animals and the absence of assistance provided to people to deal with the presence of wild life in their areas. But this doesn't help in relaxing the situation in any manner, in fact it worsens the conflict scenario. There are different views held by different set of people, some do not want the animals in there are some feel they have equal rights to live. A few of the respondents from Ahmednagar whom we had interviewed regarding the leopards presence had an answer like "Sugarcane has got prosperity to our lives and along with sugarcane came the leopard so we are accepting it as much as we accept the prosperity."

Another famous example is of Mumbai where large number of leopards use the space inhabited by one of the highest density of humans on this planet. Earlier due to fear and severe political pressure to trap leopards simply because they were sighted where we did not expect them to be, this very act of trapping led to an increase in attacks on people near the release sites. The forest department in a unique initiative tackled this problem by increasing their capacity, by increasing the knowledge base (from both scientific and traditional information) and disseminating the knowledge about leopards and the issue using the citizens of Mumbai and their staff to share the precautionary practices people need to use for their own safety. With increased understanding and better relationship with the forest department came increase acceptance and trapping pressure deceased, and so has conflict.

However, in the case of both livestock and crop loss, the farmers, especially the more marginal ones are affected seriously many a times due to wildlife damages. On the other hand you also hear of farmers in Tadoba area who say that they traditionally kept aside a fraction of their crop for wild herbivores to feed on. Also in places like Orissa, they think their harvest would be better if elephants came and fed on their crop..

In the year 2002 Maharashtra had to tackle a new problem which it had never faced before, that of wild elephants. Presence of elephants was found in Kol-

But even after all these efforts both people and elephants had to face a negative impact. Some local remedies practiced were throwing one too many fireballs at elephants, inserting live wires in the water ponds which the elephants used to drink

hapur and Sindhudurg Districts. During this time efforts were undertaken by the forest departments of Karnataka and Maharashtra for the removal of elephants. Along with the removal Maharashtra forest department also monitored the elephant herds and spread awareness amongst people to avoid conflict. But even after all these efforts both people and elephants had to face a negative impact. Some local remedies practiced were throwing fireballs at elephants, inserting live wires in the water ponds which the elephants used to drink water from. These short term remedies seldom work especially with elephants.

In one area in Tamil Nadu Dr. Anand Kumar has with the tea garden management and forest department developed a sms based early warning system so people know of the location of the elephants so they can avoid dangerous encounters and there has been a tremendous decrease in conflict. And therefore it makes much more sense to focus on increasing the safety of humans rather than pushing or removing animals.

Thus the central issue here is how different people view the wild animals and the resultant precautions they take to deal with the economic losses they have to them. Unfortunately not everyone has incorporated traditional knowledge about best practices to deal with wild animals in order to reduce losses. As the Mumbai and the Tamil Nadu example shows, there are ways of using innovative methods and community involvement to avoid these conflicts but the real challenge is to change people's perspective. The media is a major player, more often giving sensational twists to the events for increased TRP's that is detrimental to people and wildlife. The media can also, as it did in Mumbai, change from sensational reporters to acting as an agent for informing people about the precautions they can

“While the rest of the world is trying to understand how to survive with wild animals that are now re-colonising after past extermination, India has always been a country where shared spaces between high density of people and wildlife was a rule than an exception.”

take and increasing the understanding about the wild life in shared landscapes.

While the rest of the world is trying to understand how to survive with wild animals that are now re-colonising after past extermination, India has always been a country where shared spaces between high density of people and wildlife was a rule than an exception. It is in our hands to choose a right path between conflict and co-existence. Based on traditional and scientific knowledge coupled with the modern technological advancement, choosing the path of co-existence is the more peaceful option than perpetual conflict that is detrimental to the people and the wildlife.



Vidya Athreya is an ecologist who has been working on the human leopard conflict issue since 2003. She is currently with Centre for Wildlife Studies (CWS) and Wildlife Conservation Society (WCS) - India program, Bangalore.



Nikhil Surve is an ex-xavierite and a graduate from the Wildlife Institute of India. He has worked extensively on leopards, especially in the unique scenario of Mumbai, which has had his interest from the very start.

A Tern Tale

Sometimes creatures touch much beyond the minds and enter our hearts.
Experiences like these are what connect us maybe..

The trickle of visitors to the Behavioural Ecology laboratory increased with her sudden entry, and even our routine was altered by her presence. Nay! She was not some stunningly beautiful lass who wreaked havoc among hearts. She was just a bedraggled forlorn little river tern chick who, despite her nondescript appearance made her way on webbed feet into our hearts.

The day she arrived we received her gladly, out of curiosity and pity. She had been wrenched from the security of her nest and rudely dumped amidst a horde of inquisitive human beings. Initially the shock of being in a new environment kept her silent and watchful, if she could speak I guess she would say that she was utterly homesick, dismayed, frightened and sad. She was after all only 12 days old. We humans too have a streak of gentleness though rarely visible in our unholy selves, and her utter helplessness tugged at our heartstrings . We tried our best to make her feel at home in a laboratory full of cupboards, books, test tubes, formalin jars, files, balances, buckets, tables, chairs and what nots. It was a heart-rending change for a little chick habituated to rocks and pebbles, freshwater and open blue skies.

Biological theories like imprinting, adaptation, survival etc put forth years ago held true as she slowly adapted herself, to her new lifestyle. She gradually found her voice (a most beautiful and trilling call) and overcame her inherent fear of man. Her little inquisitive brain began working and soon she began to explore her new surroundings.

Whether she liked it or not is questionable, but she never displayed any aversion and was most compliant with our actions. To date she must be the only river tern to have covered 70 odd kilometres on a motorbike, stuffed in a khaki bird bag, slung across the driver's chest. Did the noise of the engine/motor intimidate her? Did she feel suffocated in the bag? Who knows? These are questions that will forever remain unanswered.

Her growth rate was tremendous and like proud parents we gloated over every gram and inch she gained. She underwent her regular growth checks with minimum fuss, occasionally emitting a woeful long call of protest. What chance had a chick weighing 160 grams have against human beings averaging 60 kgs? She kept us company when we worked into the late hours of the night. We changed her days and night at will and yet she never uttered a chirp of protest.

Some of her natural instincts surfaced even in these unnatural surrounding. Her affinity for water was amazing and we never had to trouble ourselves regarding her hygiene. As and when it pleased her she went for a dip in the little plastic tray that we had kept for her. She preened her feathers meticulously and we boasted amongst ourselves about how 'healthy' she was.

She was aggressive with outsiders, but was totally loyal to those who fed and looked after her. When we wanted to display her intelligence to all and sundry, she very willingly obliged, by going after the shells and other 'toys' we had kept for her. She was ever watchful and her beady black eyes never shut for more than a few seconds in the day.

She begged in such an endearing manner for her food, with such accuracy of timing, that one could say for certain that nature had endowed her with a very efficient biological clock.



We talked of how she had forgotten to feed and catch fish, but we did not blame ourselves for it. We had manipulated her diet and instinctively she revolted. We crushed her tiny rebellion and stuffed unsavoury food* down her fragile gullet, to prove that we were good foster parents. We stole her from her home, we gave her food, we made her change her ways and habits to suit our convenience. In return she gave us love, trust and priceless data.

Questions of her future were raised and hesitant proposals for her release were made, but deep down at heart all of us knew that we had spoilt beyond redemption. We had destroyed and crippled her knowingly. She was now a misfit in her world as well as ours.

Little tern, you have flitted and flapped your way through the laboratory managing the bends and curves with the inborn grace of your race. How will you manage this crucial turn in your life? Will you turn towards freedom or death? The decision of life or death rests on your slender wings. May they take you to heights unreachable to humans who marred your life and yet in their own selfish way loved you.

**The state of Gujarat was experiencing a drought for three continuous years during which period no one was allowed to catch freshwater fish. This was the reason for us having to feed the bird with marine fish.*



Professor Smita Krishnan is the current Head of the Zoology Department at St. Xavier's College, Mumbai and a person with many a wonderful experiences in life.



In Conversation with... Dr. Claire Spottiswoode

Dr. Claire Spottiswoode is an ornithologist and research fellow at the Department of Zoology, University of Cambridge and has been studying brood parasitism all her life. She also happens to be one of the few people who are truly trying to understand the unique interaction between Honeyguides and humans in Mozambique!

Your research work has largely been on the African Greater Honeyguide, which has a distinct relation to the Yao community in Mozambique, what caught your interest in this species and how would you describe this link between man and animal?

Greater honeyguides are the Jekyll and Hyde of the bird world – they're both brutal brood parasites of other birds, and cooperative mutualists with humans whom they guide to wild bees' nests. For many years I've studied the dark side of their lives in Zambia, involving lots of death and decomposition down dark burrows underground. But I'd been fascinated by their co-operative behaviour since I was 11 years old and heard the Kenyan ecologist Hussein Isack give an electrifying talk about honeyguides and Boran honey-hunters in Kenya in the 1980s. His ground-breaking work showed that the bird-human relationship works because the honeyguides (who eat wax) know where the bees are but can't get at the wax, whereas the humans (who eat honey of course) know how to get at the wax but can't find the bees. So it was thrilling, over two decades on, to meet my co-authors Keith and Colleen Begg who do wonderful conservation work in partnership with communities living in the Niassa National Reserve in Mozambique, and to learn from them that

honey-hunting in cooperation with honeyguides is still commonplace at Niassa. What's more, Keith and Colleen told me, honey-hunters at Niassa have a special call – “brrr-hm!” – which the hunters say attracts honeyguides and maintains their attention. This was instantly intriguing – could this really be an example of reciprocal communication between humans and a wild animal? We knew from Dr Isack's wonderful work that honeyguides definitely communicate to humans – their calls and behaviour indicate the direction of bees nests – but the Yao honey-hunters' claim implied that humans communicate to honeyguides, too. What we've done in our experiments in Mozambique's Niassa National Reserve, is to complement Isack's work by showing that humans communicate back to honeyguides, too: honey-hunters use special calls to signal to honeyguides that they're eager to follow, and honeyguides use this information to choose partners who are likely to be good collaborators. It sounds very much like a two-way conversation between our own species and a wild animal, benefiting both partners.

○○ *Yao honey-hunters' claim implied that humans* ○○
communicate to honeyguides, too.

As in case of the Honeyguide, how unique is it to find a phenomenon like this occurring in nature? Are there other examples of the same kind?

The only comparable example I'm aware of is the foraging association between artisanal fishermen and wild dolphins who herd shoals of mullets into their nets, and in so doing manage to catch more fish for themselves. This has been beautifully described from Brazil and elsewhere, starting with Pliny the Elder nearly two millenia ago!. But we don't yet know whether it is mediated by a reciprocal communication system like the honeyguide-human relationship in Africa.

Is it rare to find such relations between man and animal? They appear very different from something like domestication, what according to you is the significance of such a link?

I completely agree with your assessment that the human-honeyguide relationship is different from domestication. Strictly speaking, domestication means humans imposing some sort of genetic change on other animals – like we've done by artificially selecting for certain traits in livestock or dogs. This is definitely not the case in honeyguides – they're free-living wild animals that reproduce of their own volition (albeit by brutally exploiting other birds to do so!). As far as we know, honeyguides don't connect with humans in any other context, although they're clearly attracted to people when they're in search of wax.

In your opinion, how does your study shed light on man- animal relations?

I should emphasise that our study only showed, by controlled experimental means, what local people in Mozambique have already known for generations! To me their knowledge, together with our study, underlines that humans are continuous with the natural world, and that our interactions with nature are products of natural selection. More specifically I'm intrigued at the variation across in the calls that humans use when seeking and following honeyguides – utterly different the Yao's "brrr-hm" call made for the same purpose. For example, Dr Brian Wood from Yale has shown that Hadza honey-hunters instead use a lovely melodious whistle which they too say attracts honeyguides and maintains their attention. Ultimately, Brian and I would love to know whether human culture and honeyguide behavioural traditions coevolve with one another across Africa – but this still lies ahead.

On a personal note, how did you really get into the profession and studying something this interesting?

I come from South Africa, a place where there is still wild nature all around, and first fell in love with biology – and especially birdwatching– aged seven. I've yet to shake it off. My research in bird ecology and evolution still feels like a slightly grown-up version of that early obsession. I feel incredibly lucky to be allowed to spend most of my time answering questions I think are interesting, and sharing that with wonderful people living in various parts of Africa, as well as colleagues in the lab at the University of Cape Town and the University of Cambridge.

Do you think your work on the honeyguide can serve as an inspiration for resolution of man and animal conflict or be of use in that perspective?

The honeyguide-human relationship works because both parties get a useful payoff from it (food), and each species gets a reward as a byproduct of the other species's gain, at no additional cost. This is probably what has kept it relatively free of conflict and so evolutionary stable for so very long, and Mozambique's Yao people for example have great affection and respect for the honeyguide. But there are losers too in this relationship – the bees whose nests and food supplies are destroyed in the process, and the mature trees that are felled to gain access to the bees! I guess that this underlines further still that we humans are part of the planet's ecology, for better or worse, like any other species we share our planet with.



Above: Dr. Spottiswoode and team on field at an experimental transect.

“Mozambique’s Yao people for example have great affection and respect for the honeyguide.”

What advice do you have for people who’d like to get into this field? How has working in the field changed your outlook on life?

I think the key thing is to find something that fascinates you, and turn it into a field of study. Those of us who are interested in field biology are lucky because it’s especially easy to get involved – natural history is easily accessible everywhere and gives us the foundations for all of ecology and evolution. So I would say, find something that you really really love, find others who share your passion, and make it your career. It’s that passion that will make you look forward to going to work every day. But of course science isn’t just a job, it’s a complete life. It isn’t something you can only think about in the lab – it’s all around us, and brings the world into wonderful proximity. And above all, it’s amazingly good fun – nothing beats the thrill of discovering for the first time how some new little bit of the world works.

If the crocodile goes, we go

A tradition of crocodile protection...

Freshwater crocodiles and people who reside along rivers share a common natural environment. For people who live along rivers sheltering crocodiles, a cautious coexistence with these supreme creatures becomes a way of life. Crocodiles and humans have shared a rocky relationship since time immemorial. The ancient reptile finds itself in both extremes—that of reverence and aversion. Crocodiles have been declared as vermin in some lands and regarded as sacrosanct in other parts of the world.

In the ancient Egyptian pantheon, Sobek or the crocodile God, was the protector of the Nile in addition to being an embodiment of fertility, military competency and the ruling power.

According to a creation myth, Sobek laid eggs on the banks of the waters of Nun (watery abyss) and thus created the universe. The local people of Mauritania who live in close proximity to the West African crocodile (*Crocodylus suchus*) provide protection to these creatures. This is due to their belief that the crocodiles are tied to the water. If the crocodiles go, the water goes as well. The crocodiles live in harmony with the people, and it seems that they never attack swimmers. In Gambia, Kachikally Sacred Crocodile Pool is home to 80-90 crocodiles. The pool is supposed to have healing powers and is visited by infertile women. Politicians regularly visit the sacred pool for good luck.

Back home in India, our rivers shelter the mugger crocodile also known as the Indus, Sindhu and marsh crocodile. It is an opportunistic apex predator; it sneakily hides in mucky waters and then executes a surprise attack. Muggers are found in most rivers, lakes, swamps and marshes in India. Human-mugger conflict has been reported from various parts of the country. The main threats to the mugger are habitat degradation, drowning in fishing nets, egg predation by people and the use of crocodile parts for medicinal purposes.

Goa houses a very small population of muggers. These are spotted in mangrove habitats. Crocodiles are not native to Goa. They were introduced into the Cumbharjua-Banastarim canal by Adil Shah of Bijapur in 1510.

These muggers were used as a form of warfare against the Portuguese general Afonso de Albuquerque and his soldiers in the Banastarim Battle of 1511. These ‘fighter’ crocodiles are revered by a few families in Goa in a unique ritual known as mannge thapnee in Konkani. Mannge means ‘crocodile’ in Konkani. The ritual is celebrated in Durbhat (a village on the banks of the Zuari River) every year on Paush Amavasya in the Hindu calendar, which corresponds to the first new moon day in January.

Charan Desai, a wildlife rescuer and founder of Study and Awareness of Wildlife and Environment (SAWE), interviewed two locals from the village Talavi (a neighbouring village to Durbhat,). Tukaram Mulik said that he had been performing mannge thapnee for the last thirty years, after the death of his father. He performs this ritual along with thirty other people from fishing and farming families. The main purpose for this practice is to please the crocodiles in order to keep the paddy fields safe from any crocodile visits. In the present times, there have been no reports of crocodile attacks in Goa.

Tukaram Mulik said that he had been performing mannge thapnee for the last thirty years, after the death of his father.

He performs this ritual along with thirty other people from fishing and farming families.

Mulik emphasizes that they feel safe after conducting the ritual and are able to traverse across fields and bundhs without any fear. A clay replica of the crocodile is created in the khazan or saline floodplain. The clay mannge is decorated with shells, flowers, vermilion and incense sticks. The men in the family conduct the prayers. They practise this ritual every year without fail. A chicken hatchling and an egg are buried inside the mud crocodile as sacrifices along with puffed rice, coconut and jaggery.

This instance of human-animal coexistence is an example that can be adapted to other regions of India. One can only hope that traditions such as this one supplements conservation efforts to protect threatened species and their habitats in the wild. Can these examples from remote India inspire our urban educated youth to lead an environmentally conscious life?

Malaika Mathew Chawla, SYBSc, Parvatibai Chowgule College of Arts and Commerce, Margao.



The Spirits Within...

Finding how we associate our lives with creatures other than ourselves, each one to their own...

Shamanism is an ancient healing tradition and a way of life for some people, it's practices and components differ from culture to culture but they usually revolve around altered states of consciousness. It is also a manner in which more often than not, humans have understood their relationships with animals.

The relationship between a shaman and his animal companion is starkly juxtaposed to that of modern, urban society in which animals are seen as inferior and subservient to humans. According to Shamanic belief however, everything is alive and has equal power and wisdom, animal spirits in fact add to the power of Shamans and are essential for success in any venture undertaken by them. In Shamanism animals are seen as spirit guides, also known as spirit or power animals, every human is said to have one such 'guardian angel'. These spirit animals choose the shaman they wish to accompany while offering them insight, guidance and protection throughout their 'vision quest'.

A person may have several guardian spirit animals over the course of their life, beginning from childhood, if for example, a power animal leaves and another doesn't come to take its place, that particular individual is believed to be disempowered, vulnerable to illness and bad luck. Among Native Americans, a person who loses their spirit animal is said to go to a shaman, who then undertakes a journey to the spirit world in search of a new spirit animal.

Even though I keep referring to spirit 'animals' - plants, trees, reptiles, insects or sea creatures can be guardian spirits as well, with the exception of domesticated animals - seeing as they are already of 'service' to humans. Each animal is said to impart their own wisdom and knowledge to the shaman, for example, a hawk spirit will give specific wisdom and lend hawk like attributes to the person. These 'functions' of the power animal are closely related to cultural beliefs and practices for example, the Tlingit of Southeast Alaska believe that Orcas were created from yellow cedar, by a hunter, in order to kill his wicked in laws, after their death, the hunter ordered the Orcas to never harm humans again, thus the Tlingit consider the Orca a benevolent custodian of the sea.

Shamans believe that different animal guides come in and out of our lives depending on the direction we are headed, the experiences we had, are having or

will have, there is however only one totem animal who acts as the 'main guardian spirit', both in the physical and spiritual world, these spirits appear to humans in dreams and meditations and could possibly manifest physically for Shamans.

The belief in a guardian animal originated from the concept of animism which believes animals possess souls and consciousness which exists apart from their bodies. Animals in Shamanism are not only restricted to their roles of spirit guides, shamanism also involves human- animal hybridity, which is, as if the soul of a shaman and animal merged together. For Native American shamans the 'visual' and 'visceral' impact of their trances lies in this experience of 'becoming a single animal', several animals or 'watching animals change into other animals and beings'.

The leader of the Quiche people for example identifies as a snake, eagle and 'tigre' (jaguar), similarly a person from Santiago claimed he became a tiger, who approached 'an animal, any animal, saw its neck and then experienced what a tiger feels like when looking at its prey'

In this case it's almost as if the shaman lose their individual identity and take up that of the animal, these visions are often 'terrifying and distressful', for example Lewin (1931) describes such a feeling of being squeezed to death in his vision by huge brightly coloured boa constrictors such as those usually found in the rainforests of Panama. Visions of flying in an avian form have also been described which are said to lead to a powerful feeling of emotional unity and the recognition that all beings are truly alive.

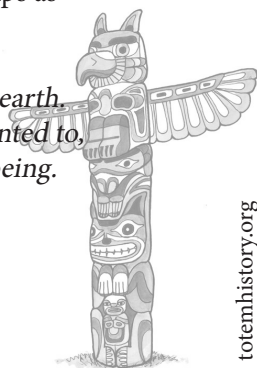
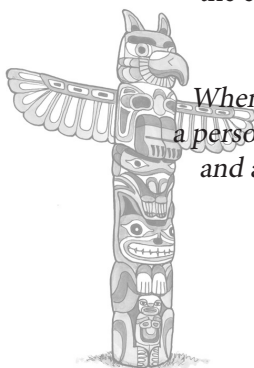
Conclusively the lyrics of the Netsilik Inuit song encapsulate the essence of human animal relationships as

depicted in Shamanism:

"In the very earliest time

*When both people and animals lived on earth,
a person could become an animal if he wanted to,
and an animal could become a human being.*

*Sometimes they were people
and sometimes animals
and there was no difference
All spoke the same language"*



totemhistory.org

Story by Shainya Shetye, SYBA, St. Xavier's College - Mumbai

PHOTO OPS



Above: Gulls Portraiture by Shreya Dimri, SYBSc, St. Xavier's College, Mumbai

Above on Right: A jumping spider as pictured by Shreya Dimri, with all the 8 eyes in focus.

Below on Right: An action shot of a Rosy Pelican landing in the water at The Delhi Zoo by Harshit Singh, TYBSc, St. Xavier's College, Mumbai





PHOTO OF THE SEASON

By Pushkar Sarnaik, SYJC, St. Xavier's College, Mumbai

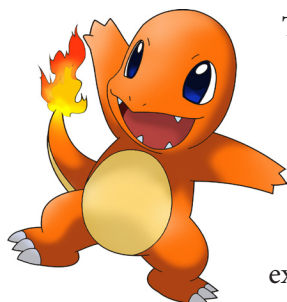


Hemkund Sahib (Chamoli, Uttarakhand): Situated 14500 feet above sea level this lake next to the Hemkund Sahib Gurudwara is used for holy baths by Sikhs. bone chilling temperature and minimum oxygen level makes route a very dangerous but still huge number of people visit throughout the summer.

Welcome to the Land of Pokémon

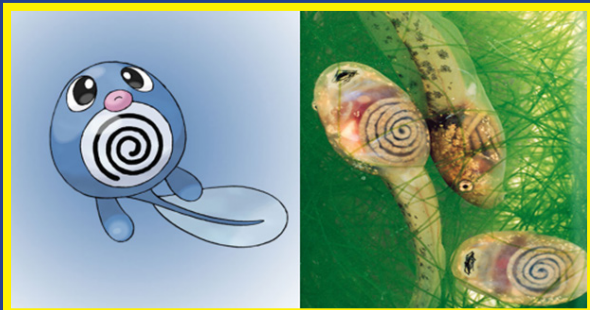
Pokémon, a worldwide phenomenon that arose in 1996 is still of huge appeal to a very wide audience. These amazing creatures have been the base upon which I built my childhood, and as I grow older, my love for them grows as well. With the ever increasing number of game and the addition of newer and newer Pokémon, GameFreak does their best to adopt designs from every field of life. Obviously, during the creation of this magnificent world, a lot of inspiration was taken from real life (I don't mean 3 headed dragons or Mutated alien DNA). I shall be your guide through this journey, in an attempt of joining the world of Pokémon and our own!

Your trusty Charmander and Salamander



There are many cases where we see Pokémon and find a distinct relation to their real life counter-part. Like Squirtle is a tiny turtle, or Victreebel is a pitcher plant, but many people question as to why a salamander gains a fire typing at all? Morphologically it is evident that Charmander shares a very similar appearance to a salamander, and you would expect it to possess a water typing, but this fire lizard actually takes its inspiration because of the behaviour of salamanders during medieval England. Being a poikilo-

therm, salamanders desperately looked for warmth, and it found warmth in the form of burnt logs in the fire place. After the fire dies out several of these salamanders scurry into the ashes to shield themselves from



L: Poliwag and the Translucent Tadpoles are another example of animal inspiration found in Pokemon. The whirled pattern of the digestive tract is present in both!

the cold. The “Char” part of charmander arises when the fire place was used again. Humans unaware of the presence of the salamanders again light wood for warmth. This causes the poor amphibian to run out. During its flight its tail would often catch fire and the people called them Fire Lizards as they believed the salamander was born from the fire.

Trapinch and the Antlion



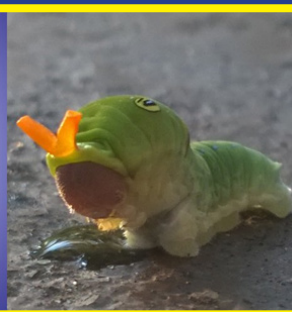
Antlion, an arthropod of class insect is well known for its aggressive larval stage, rather than for its adult form that looks like a dragon fly. Antlions have an incredibly unique method of hunting. What they do is, dig shallow pits in the sand and sit at the bottom. Any ant foolish enough to even come close to one slides in and is unable to escape because of how loosely the sand is held. Every Pokémon possesses a specific ability that helps it in battle in the wild (Predation in the Pokémon world). Trapinch has the ability of Arena Trap. This ability prevents any Pokémon from escaping. Trapinch on evolving forms Flygon. A universally loved dragon type which is similar to the post cocoon adult form of the antlion.



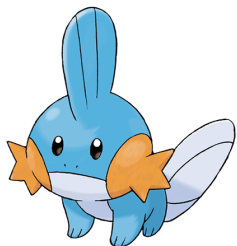
The Man who Made it

Meet Tajiri Satori, best known as a Japanese video game designer and the founder of the company Game Freak. As a child, Tajiri enjoyed insect collecting as a hobby, which would be an inspiration for his later video game work. Other children called him “Dr. Bug”, and he wanted to become an entomologist. As the urban areas of Japan spread and more land was paved over, habitats for hunting bugs were lost. Tajiri wanted his games to allow children to have the feeling of catching and collecting creatures as he had.

R: Caterpie and The Eastern Tiger Swallowtail Caterpillar are exact replicas. Not just are their looks alike but the habits, habitat and natural history of both the subjects are the same!



Mudkip and Axolotl

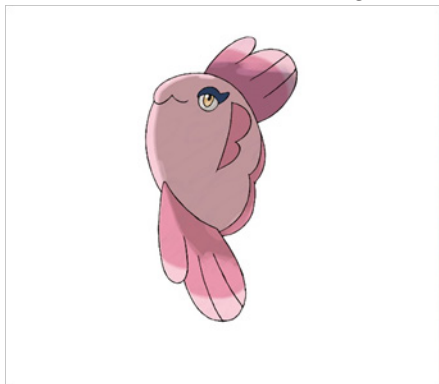


fish, the axolotl has 4 small appendages, just like mudkip! But the most distinguishing similarity is the presence of the external gills in both Mudkip and Axolotl. The axolotl almost entirely exhibit neoteny and hence do not lose the external gills during metamorphosis (Just like Mudkip even during Pokemon evolution)



The Mola Mola fish and Alomomola.

The MolaMola fish is incredibly huge and looks like it belongs in a Disney movie. Having an almost flat body, this fish can weigh between 250-1000 KG! Mola Mola fish are found almost exclusively in tropical waters, suffering disorientation or even death when subject to lower temperatures. It is often seen floating near the surface, soaking in rays before diving deeper. They live almost exclusively on a diet of jellyfish. Alomomola, even shares a similar habitat, feeding habit, etc. Alomomola, is found solely in the water of the more tropical regions of Unova and Hoenn. Even in the game these Pokémon are found exclusively in areas where you also find Frillish and Jellicent, the Jellyfish Pokémon. Kudos to GameFreak for attaining this level of detail into a single Pokémon!



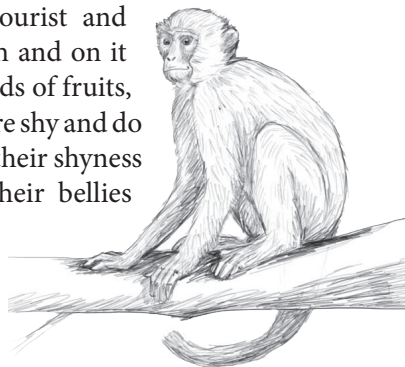
Like the Mad Hatter's Tea Party?

About Hundred and fifty kilometers north- east of Bangkok lies the City of Lopbhuri dubbed as the Monkey City. Here one can attend the “World’s Wildest dinner party” that is the Monkey’s Buffet festival. The city is known for the hundreds Monkeys living here. Crab-Eating Monkeys (*Macaca fascicularis*), also known as long tailed macaques are the native population living here. Since pre-historic times they have been living alongside humans.

The residents of Lopbhuri organize an annual festival for these untamed and crazy co-inhabitants. This festival is one of the strangest festivals in the world. It takes place on the last Sunday of November. On this day the monkeys are offered fruits, vegetables and deserts. This festival was started in 1989 by a local businessman Yongyuth Kitwatananusont. In Lopbhuri culture monkeys are revered and considered as the descendants of the heroic deity Hanuman. According to folkfare, Hanuman helped Lord Rama in rescuing his wife Sita from the Demon. In Lopbhuri culture Monkeys are considered to bring good luck and fortune.

“In Monkey’s we see an exaggerated version of ourselves and are reminded that no matter how different we are, tolerance brings good Fortune to all”.

This festival was originally started to attract tourist and increase tourism. Tables are spread with red cloth and on it are placed nearly 2000 Kilograms of food- pyramids of fruits, vegetables and nuts. These simian kleptomaniacs are shy and do not come to eat the food. Eventually they get over their shyness and jump on the tables gorging and stuffing their bellies with the tasty food available. Once their bellies are full, then begins the actual fun. Food is thrown around, soda at each other and these beautifully decorated tables then become a mess something like at the Table at the Mad hatter’s Party in Alice in Wonderland. On other Days, the monkeys can be seen roaming around the city in gangs and playing pranks and stealing food from unsuspecting tourist. These monkeys cause a lot of destruction to properties but are still pampered and fed with food.



FANTASTIC BEASTS

And where they found them!

What's common among dragons, acromantula, hippogriffs and thestrals?

They all exist in the world of magic, and they all have their inspirations deeply rooted in the animals muggles know all too well.

Let's take a look at how some of our favourite magical creatures in the book and film were inspired by the ones in existence.

THE NIFFLER



The creature that stole everyone's heart in the movie was the 'Niffler'. It is a black-furred creature, the size of a platypus. It even has a beak like a platypus'. What's unusual about it, is that it has a pouch on its belly, quite like a kangaroo's, where it stores shiny objects. The pouch is unbelievably large on the inside. The Niffler's attraction to shiny objects reminds us of a magpie's supposed love for anything lustrous. The inspiration for the Niffler's chaotic persona was a video on the internet of a honey badger raiding a

house for food, says Christian Manz, a visual effects supervisor for the movie. They created the Niffler with looks similar to that of a honey badger, and gave it some traits of other animals as well.

THE BOWTRUCKLE



The word 'Bowtruckle' can be translated to 'a dwelling limb of a tree', based on old dialects in English and Scottish. About the size of a human hand, Bowtruckles are plant-like life forms. They are green, have four legs that look like plant roots, two hands with three or four root-like fingers each, and a head that sprouts leaves instead of hair. One can safely say that they were inspired by the camouflage abilities of stick insects.

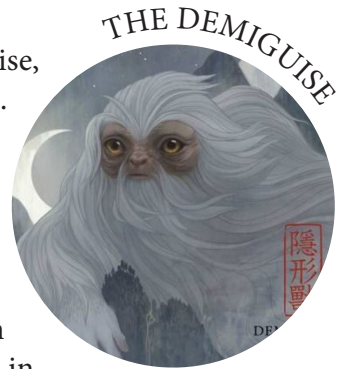
Newt's most memorable retrieval of a magical creature is definitely that of his Occamy. An Occamy is a blue, serpentine creature, with some features of a tropical bird, like its beak for instance. Occamies possess the ability to change their size in order to fit into available spaces, quite like gaseous substances. It seems that the looks of the Occamy were directly inspired by a hyacinth macaw, which is found in northern Paraguay. It is primarily blue in

colour, and has yellow eyes, just like an Occamy. It is known that people sometimes capture young tropical birds, like the hyacinth macaw, stuff them into small cages, and proceed to ship them internationally. This is probably why the Occamy was given the luxury of being able to fit into tiny spaces. Also, exotic birds fetch a huge amount from any international buyer. This might be why Rowling wrote that Occamy egg shells are pure silver, and thus, can help you earn a lot of money.



All of us remember the wise-looking Demiguise, a primate with long, silver hair and huge eyes. The Demiguise is a cross between an orangutan and a sloth. Tim Burke from the VFX team said that he looked at the work of an artist called 'Veruschka', who paints people that blend in with the background or the environment, and that led to the explanation for the Demiguise being able to make itself in

visible. The truth is that its fur can change colour to blend in with the surroundings. It's like a next-level camouflage. Demiguise fur is also used to make Invisibility Cloaks, which lose their invisibility after being used for a while. These aren't like Harry's Cloak, which is a Deathly Hallow, and never loses its properties. There are many more creatures this kind in the film but the reality is that when you read up on fictional versions like these you come to realise the many creatures that actually exist and our fantastic beasts themselves!



Internships Galore

Multiple students from the department interned at various places this year, here's a look at where all they've been.



Vinsea interned in the Summer Research Fellowship Program conducted by the Indian Academy of Sciences. She designed her own experiment under her mentor to study the mating behaviour of paper wasps *Ropalidia marginata* (Eusocial wasps whose nests are made of paper).



Lisa interned at Phoenix Veterinary Specialty, a small clinic located next to Prabhadevi Temple under Dr. Shiwani Tandel in April 2016. Dr. Shiwani specializes in exotic animals and is a renowned avian veterinarian and hence Lisa has fond memories of helping treat turtles, exotic birds, guinea pigs, babblers along with Chatur the rooster and Artemis Beth the fruit bat.



In the Summer of 2016, Karan worked under Dr. Samuel Hung, Director of Hong Kong Dolphin Conservation Society. He studied techniques involved in cetacean research and on conservation of the Indo-Pacific Humpback Dolphins.

Dolsy and Sandra (SY students), along with three other Second Year and seven

Third Year students worked every Sunday at Sasoon Docks early in the morning carrying out a census on Shark population - Species, sex, size etc. Over the different seasons, they came across many different species of Sharks from Dog Sharks to hammerheads to Blacktips, all in varying sizes.





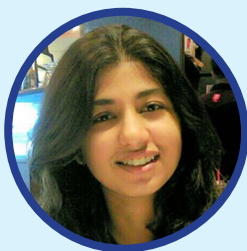
Udit Nair and Harshit Singh worked for a month in Delhi as part of The Black Kite Project, a collaborative effort of WII and The University of Oxford to study the unusually large population of Black Kites in the city and how their population ecology functions. The work involved studying various different nests of the species, carrying out



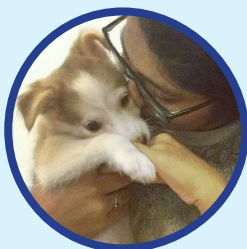
measurements, analysing data, setting up camera traps, radio tagging the birds, etc.



Shivira Shukla interned for the Wildlife Institute of India, as a field intern based in Uttarakhand, and was part of a project for the continuous monitoring of avifauna populations during the breeding season. Her tasks included walking transects and doing point counts along with spot mapping and doing a comparative analysis with the previous years data.



In the winter of 2016, Harshini Jhala interned with National Trust for Nature Conservation (Nepal). She was involved in conducting household surveys for local communities around Chitwan National Park to better understand the role that the local communities play in conservation. She also participated in Individual ID based monitoring of Rhinos in Chitwan.



In the summer of 2016 Marishia Rodrigues interned with the Wildlife Institute of India, at Pench Tiger Reserve, Madhya Pradesh on the project "Predation ecology of large carnivores". For density estimations of predators, camera traps were used to identify individuals. Attitude surveys were conducted to estimate people's perception towards wildlife.

THE STRANGE EXISTENCE OF VELVET WORMS

Discovery of electricity, gravity or advancement in technology might be the greatest achievement. An even greater achievement, however, was made by much smaller bizarre creatures, in a bid to escape the pressure of living in the increasingly hostile seas. They sneaked ashore about 480 million years ago. Life in seas at the beginning of Cambrian period was more like hunger games. Creatures that had best adaptations to fend off predator, thrived successfully like Trilobites, they had skin like armour plates.

Amongst the strange and easily misunderstood was the marine caterpillar like creature called Hallucigenia. On its underside of wormlike segmented body were many long stiff legs while on topside it had soft brush like tentacles for catching prey in the water. At least that was what everyone thought. A Swedish expert discovered that the scientists were looking it upside down! It actually had those soft tentacles for moving around the ocean floor and the stiff long structures for defence. Due to intense competition in Cambrian seas, they gradually started evolving. They took one step forward by starting new life on the land. About 480 million years ago, the Hallucigenia were in process of evolving into the great grandparents of today's terrestrial velvet worms.

They had most dramatic evolutionary changes. Back then they had extra-terrestrial looks but through the passage of time they wriggled onto the land to be today's beautiful bright coloured velvet worms. Each segment is equipped with well-adapted pair of legs. The velvet worms possess a weird trait that sounds like sci-fi. These creatures can spray a gluey slime up to 30 centimetres (1 ft.) away to immobilize prey. The scientists were surprised to discover that the spray is directed by the movement of fluid through the two legs adapted into two openings called oral tubes, rather than by muscles. It possess stub feet which is without any joints. They show hydrostatic locomotion. They have sharp fangs hidden under soft mouthparts, can inflict a nasty bite to humans. Using the long antennae extending from its head, the velvet worm typically investigates its potential prey with a gentle tapping.

More than 150 velvet worm species have been recorded to the date. They have thought to share a connection with arthropods and annelids. Today they are on verge of extinction due to deforestation and encroachment of human population into dense forests. Only time will tell whether these intriguing creatures can re-kindle their need for evolution, in order to save their ancient ancestry.

Mrunal Deolalkar, FYBSc, St. Xavier's College, Mumbai



References

If the crocodiles go we go by Malaika Chawla

Banerjee, A. (2016, January 8). A prayer for the crocodile. Retrieved December 20, 2016, from Live Mint: <http://www.livemint.com/Politics/XT3XAh4KoIjxWqOY4dgwcO/A-prayer-for-the-crocodile.html>

Choudhury, B., & de Silva, A. (2013). *Crocodylus palustris*. Retrieved December 28, 2016, from The IUCN Red List of Threatened Species: <http://dx.doi.org/10.2305/IUCN.UK.2013-2.RLTS.T5667A3046723.en>

James, J. (2012, September 13). The crocodile feeder of Ivory Coast. Retrieved December 28, 2016, from BBC News: <http://www.bbc.com/news/magazine-19576296>

Mayell, H. (2002, June 18). Desert- Adapted Crocs Found in Africa. Retrieved November 28, 2016, from National Geographic : http://news.nationalgeographic.com/news/2002/06/0617_020618_croc.html

Millmore, M. (2016). Kom Ombo Temple. Retrieved December 26, 2016, from Discovering Egypt: <https://discoveringegypt.com/pyramids-temples-of-egypt/kom-ombo-temple/>

The Spirits within by Shainya Shetye

<https://www.shamanlinks.net/shaman-info/the-spirit-world/power-animals/http://www.animalspirits.com/index1.html>

<http://psychiclibrary.com/beyondBooks/animal-spirit-guides/>

IN CONVERSATION WITH Dr. Claire Spottiswoode
www.africancuckoos.com

WELCOME TO THE LAND OF POKEMON
www.wikia.com

LIKE THE MAD HATTER'S TEA PARTY
holidaybays.com/monkey-buffet-festival-experience-thailands-unique-festivity
everfest.com/e/lopburi-monkey-banquet-lopburi-thailand
lonelyplanet.com/thailand/lopburi-province/lopburi

All photographs and illustrations have been photographed or created by the design team of the magazine or given by the authors themselves, except wherever mentioned otherwise

A special thanks to all the people who created the
interesting fillers for the magazine:

Anish Ittoop from SYBSc for
“Welcome to the land of Pokemon”

Pratyush Panshikar from SYBSc IT for
“Fantastic Beasts and where they found them”





Diploma in Forensic Science and Criminal Law

Our Objective is to provide an introduction and orientation to the functioning of the Indian Criminal Justice delivery system, and the role forensic science plays in this process. With the use of case studies, mock crime scenes and various practicals, it gives the students the opportunity to hone the basic skills necessary for investigation. This course is apt for all interested professionals from the fields of Police, Security, Legal Management, Media, Sociology as well as any citizen or student keen to learn more about forensic science and the law.

Duration and Timings: One Academic year (July to April)

Cost: Course Fees: -Rs. 24,000/- + Processing Charges: - Rs. 200/-

Online Certificate Program in Forensic Science

Our Objective: is to make quality forensic education easily accessible by facilitating distance learning for students interested in forensic science, living in different parts of India, with limited funds and / or access to good college level forensic programs.

Duration and Timings: 5 months (June to October or November to March)

Cost: Course Fees: -Rs. 15,000/- + Processing Charges: - Rs. 500/-

Contact us for queries

Dr. Pushpa Sinkar: pushpa.sinkar@xaviers.edu; (022)22620662, ext 355

Ms. Riva Pocha: riva.pocha@xaviers.edu

Ms. Sejal Gupta: sejal.gupta@xaviers.edu

Ms. Alethea Vaz: alethea.vaz@xaviers.edu; (+91) 9867320768

